

Functional Document -- IBMS (Integrated Business Management Suite)

Field	Detail
Project Name	Integrated Business Management Suite (IBMS) v2.0
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1. Project Overview

IBMS is an AI-first Enterprise Resource Planning (ERP) platform built with **FastAPI** (Python). It provides real-time business dashboards, AI-powered analytics, risk scoring, fraud detection, compliance checking, budget optimization, and lead scoring -- all exposed via a REST/WebSocket API with a responsive single-page frontend.

1.1 Objectives

- * Deliver a unified business intelligence dashboard with real-time KPI tracking.
- * Provide AI/ML-powered services for forecasting, anomaly detection, and fraud prevention.
- * Enforce enterprise-grade security (JWT auth, TOTP 2FA, RBAC, CSRF, device fingerprinting).
- * Support multi-tier deployment (Docker, Kubernetes, AWS).
- * Enable real-time collaboration via WebSocket push updates.

2. Functional Requirements

2.1 Authentication & User Management

ID	Requirement	Status
FR-01	User registration with email and password	Completed
FR-02	User login with JWT access + refresh tokens	Completed
FR-03	TOTP-based Two-Factor Authentication (setup, confirm, disable)	Completed
FR-04	Role-Based Access Control (super_admin, admin, manager, analyst, viewer)	Completed
FR-05	Device fingerprint binding on login	Completed
FR-06	Password strength validation with policy enforcement	Completed
FR-07	Silent token refresh with rotation	Completed
FR-08	CSRF token generation and validation	Completed
FR-09	Audit logging for all auth events	Completed
FR-10	OAuth provider configuration endpoint	Completed
FR-11	Brute-force protection with IP-based rate limiting	Completed

2.2 Dashboard & KPI Module

ID	Requirement	Status
FR-12	Real-time KPI dashboard (revenue, margin, risk, compliance)	Completed
FR-13	KPI history with trend data points	Completed
FR-14	WebSocket-based live KPI push updates	Completed
FR-15	Auto-refresh KPI snapshots every 15 seconds	Completed
FR-16	Company-level KPI filtering	Completed

2.3 AI & Analytics Module

ID	Requirement	Status
FR-17	Sales forecasting with confidence interval bands	Completed
FR-18	AI-generated business insights (anomalies, trends, predictions)	Completed
FR-19	Active anomaly detection on transaction volume	Completed
FR-20	Natural language AI Copilot for business queries	Completed

2.4 Risk & Fraud Module

ID	Requirement	Status
FR-21	Transaction risk scoring (0-100) with factor breakdown	Completed
FR-22	Isolation Forest-based fraud detection with review flag	Completed
FR-23	Composite risk score (amount 50%, behavior 30%, compliance 20%)	Completed

2.5 Business Operations Module

ID	Requirement	Status
FR-24	Budget optimization with growth targets	Completed
FR-25	Compliance control-set evaluation (high-value approval checks)	Completed
FR-26	Dynamic pricing recommendation (demand/stock/competitor)	Completed
FR-27	Risk-based workflow decision engine (approve/review/reject)	Completed
FR-28	Lead scoring (55% engagement + 45% fit)	Completed

ID	Requirement	Status
FR-29	Digital twin operational simulation	Completed
FR-30	Inventory reorder point prediction	Completed

2.6 System & Monitoring

ID	Requirement	Status
FR-31	Health check endpoint with Redis and uptime status	Completed
FR-32	System metrics (requests, errors, WS connections)	Completed
FR-33	Detailed system status with top endpoints and cache state	Completed
FR-34	Notification queue management	Completed
FR-35	Event stream with publish-subscribe routing	Completed

3. API Endpoint Summary

The application exposes **70+ API endpoints** across the following categories:

3.1 Authentication Endpoints (14 routes)

Method	Endpoint	Auth Required	Description
POST	/api/auth/login	No	JWT login with device fingerprint + 2FA
POST	/api/auth/register	No	New user registration
POST	/api/auth/refresh	Cookie	Refresh token rotation
POST	/api/auth/logout	Bearer	Revoke tokens, clear session

Method	Endpoint	Auth Required	Description
GET	/api/auth/me	Bearer	Current user profile + permissions
GET	/api/auth/csrf	Bearer	Generate CSRF token
POST	/api/auth/2fa/setup	Bearer	Initialize TOTP 2FA
POST	/api/auth/2fa/confirm	Bearer	Verify and enable 2FA
POST	/api/auth/2fa/disable	Bearer	Disable 2FA
GET	/api/auth/roles	Bearer	List roles with permissions
POST	/api/auth/password-strength	No	Validate password against policy
POST	/api/auth/token	No	Legacy token endpoint
POST	/api/auth/validate	No	Legacy token validation
GET	/api/auth/oauth-config	No	OAuth provider configuration

3.2 Dashboard & KPI Endpoints (3 routes)

Method	Endpoint	Auth Required	Description
GET	/api/dashboard	No	Real-time KPI snapshot
GET	/api/dashboard/history	No	Historical KPI trend data
WebSocket	/ws/kpi	Token (query)	Live KPI push updates

3.3 AI & Analytics Endpoints (4 routes)

Method	Endpoint	Auth Required	Description
POST	/api/forecast	No	Sales forecast with CI bands
GET	/api/ai/insights	No	AI-generated business insights

Method	Endpoint	Auth Required	Description
GET	/api/ai/anomalies	No	Active anomaly detection results
POST	/api/copilot/ask	No	Natural language business queries

3.4 Risk, Fraud & Business Endpoints (7 routes)

Method	Endpoint	Auth Required	Description
POST	/api/risk/score	No	Transaction risk scoring
POST	/api/fraud/detect	No	Isolation Forest fraud detection
POST	/api/risk/composite	No	Multi-factor composite risk
POST	/api/budget/optimize	No	Budget allocation optimization
POST	/api/compliance/check	No	Control-set compliance evaluation
POST	/api/pricing/suggest	No	Dynamic pricing recommendation
POST	/api/decision/evaluate	No	Risk-based workflow decision
POST	/api/leads/score	No	Lead qualification scoring
POST	/api/twin/simulate	No	Digital twin simulation
POST	/api/inventory/predict	No	Inventory reorder prediction

3.5 System Endpoints (5 routes)

Method	Endpoint	Auth Required	Description
GET	/api/health	No	Health check with uptime stats
GET	/api/metrics	No	System metrics snapshot
GET	/api/system/status	No	Detailed system status

Method	Endpoint	Auth Required	Description
GET	/api/notifications	Bearer	Notification queue (paginated)
GET	/api/audit/log	Bearer + RBAC	Audit event log (audit.view)
GET	/api/endpoints	No	List all API routes

4. User Roles & Permissions

Role	Permissions	Inherits From
super_admin	Full access (*)	--
admin	dashboard, users, reports, settings, api, ai, risk, compliance, audit	--
manager	reports.export, risk.manage, compliance.manage, budget.approve, leads.manage, pricing.manage	analyst
analyst	reports.view, ai.view, risk.view, compliance.view, forecast.view, copilot.use	viewer
viewer	dashboard.view, kpi.view	--

5. Frontend Features

Feature	Description
Login / Register UI	Auth overlay with username/password, TOTP 2FA field, password strength meter

Feature	Description
Real-Time Dashboard	KPI card grid with live WebSocket updates, animated counters
Forecast Chart	Chart.js integration with confidence interval visualization
Sidebar Navigation	11 pages: Dashboard, Analytics, AI Insights, Risk & Fraud, Compliance, Pricing, Budget, Leads, Events, Health, Audit
Theme Toggle	Dark/light mode with persistent state
Toast Notifications	Max 5 concurrent, 5-second auto-dismiss
Token Management	Silent refresh, auto-logout, CSRF injection
WebSocket Connection	Auto-reconnect with exponential backoff, health indicator
3D Tilt Effects	GPU-accelerated animations on KPI spotlight cards
Reduced Motion Support	Respects prefers-reduced-motion media query
Mobile Responsive	Collapsible sidebar, responsive grid layout

6. Non-Functional Requirements

Requirement	Implementation
Performance	15-second KPI refresh cycle; Redis caching with in-memory fallback
Security	JWT + 2FA, RBAC, CSRF, rate limiting, device binding, audit logging
Availability	K8s 3-replica deployment, health/readiness probes, auto-restart
Scalability	Stateless API, Redis cache, K8s horizontal scaling
Observability	Trace IDs on all requests, metrics endpoint, response timing headers
Compliance	Audit log, compliance engine, control-set evaluation

7. Technology Stack

Layer	Technology
Backend	FastAPI (Python 3.11), Uvicorn ASGI
Frontend	Vanilla JavaScript, HTML5, CSS3, Chart.js
Authentication	JWT (HS256), TOTP (pyotp), bcrypt/PBKDF2
Database	MariaDB 10.6 (via Frappe ORM), Redis 7 (cache)
Containers	Docker, Docker Compose, Kubernetes
Infrastructure	Terraform (AWS: VPC, RDS, ElastiCache)
Reverse Proxy	Nginx (Alpine)
AI/ML	Prophet (forecasting), Isolation Forest (fraud)
CI/CD	GitHub Actions, GitHub Container Registry (ghcr.io)

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